DecisionGuard: SME Finance Copilot. Technical documentation

Deliverable 01: architecture, core algorithms, security design

Competition	2026 Shenzhen International FinTech Competition (FinTechathon), International Track
Topic	**E: SME Finance Copilot** (financing-option matching, cash-flow management with alerts, business financial analysis)
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Date	20 Oct 2026 (final submission)
Code	`main` at the code-freeze commit named in `docs/submission/evidence/README.md`

How to read this document

Every part of the system has one of four labels:

- **Built**: merged, tested, and in this snapshot.
- **Partial**: some milestones merged; the gaps are named.
- **Not in this snapshot**: designed and specified in `docs/packages/WP-*.md`, not merged. We describe the design briefly and claim nothing about it.
- **Cut**: dropped on 19 Oct to fit the deadline (`docs/FEATURES.md`, note at the top); kept as roadmap only.

Every number in this document comes from a file in the repository, and the source path is written next to it. Worked examples were computed for this document by running the repository's own engines on the synthetic data (commands in Appendix A). Test counts are not repeated here: see `docs/submission/evidence/test-counts.md`, which `npm run evidence` regenerates.

All business data in this snapshot is synthetic. No real shop data, real users or live LLM traces are included or claimed.

Snapshot status

Area	Work package	Status	Notes
Scaffold, 58 frozen contracts (v1.2.0), synthetic universe, CI	WP-00	Built	`backend/app/contracts/`, `data/synthetic/`
Typed Fact Ledger, provenance, `/api/facts`, `/api/explain`	WP-01	Built	
Ingestion: bKash, Nagad, bank, ledgers, Bangla	WP-02	Built	

Area	Work package	Status	Notes
normaliser, OCR drafts, preview/commit			
Privacy: detectors, token vault, anonymiser, fail-closed gate, shop-file anonymiser	WP-03	Built	Called by the WP-10 gateway on every model request
Cash engine: forecast, Monte Carlo, gaps, alerts, levers, buffer, backtest, API	WP-04	Built	
Financing: true cost, catalogue, eligibility, ranking, lender score, fit on the WP-04 band, API	WP-06	Built	`fit.no_forecast` only when the cash engine abstains (too little history)
Analysis: statements, 13 ratios, expected vs reality, scenarios, stability score, API	WP-07	Built	PR #24
Liquidity mesh: netting, consent, settlement, residual routing, attestations, anchoring, API	WP-08	Built	PR #26
LLM gateway, agents, checker, critic, agreement, self-loops, abstention, copilot API	WP-10	Built	PR #29, 9/9 milestones
Security: auth, roles, tiers, approvals, audit chain, injection guard, hardening	WP-11	Built	
Trust Passport, sharing, verification, state adapter	WP-13	Partial	Lender report cut. Sharing records the owner's consent (step-up, expiry, revoke); the lender reads through a consented lender-access account, not a share link
Web app (12 milestones)	WP-14	Built	Screens whose backend is missing are listed in §2.3
ML notebooks	WP-17	Partial	Injection classifier not deployed; GPU run and market backtest cut
Docker, Caddy, deploy with rollback, backups	WP-18	Partial	China check pending; Vercel mirror cut
Receivables: aging, risk score, reminder plan, API	WP-05	Built (reduced)	Milestones 1, 2, 4, 7 (PR #31); limits, dispute log, early-pay calculator cut
Message drafts: EN/BN templates, drafts from the reminder plan, owner approval, send log	WP-12	Built (reduced)	PR #34; sending channels cut: the owner copies the approved text
End-to-end flows (12, desktop + phone, Playwright with axe)	WP-15	Built	Run in CI on every PR against the local stack; hosted run waits for the server

Area	Work package	Status	Notes
Evaluation harness (62 tasks, step logs, evidence bundle)	WP-16	Built (reduced)	PR #33; mutation score and nightly audit drill cut
Market: TCB Dhaka prices, alerts, vendor ranking, negotiation (N3, N19)	WP-09	Built (reduced)	Re-included 20 Oct. Prices are TCB's published daily lists (10 lists saved, not fetched on a schedule); vendor quotes are generated for the demo shops' own ledger suppliers (synthetic); no quote form for self-service shops
Nightly audit drill (P8)	WP-16	Cut	

Source: `docs/PLAN.md` §8 and the merged code. Statuses are re-checked at code freeze.

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1. Problem and design principles

A small Bangladeshi shop sells on credit (*baki*), takes most payments through bKash and Nagad, keeps a paper ledger, and borrows informally because it cannot show a lender a clean record. The owner needs three things from a copilot (Topic E): which financing to take and what it really costs, when cash will run short and what to do about it, and an honest picture of the business.

The theme is *AI as a Financial Participant*. An agent that acts on money must be right about numbers and must never act alone. Research summarised in `docs/CONTEXT.md` §4 shows why plain LLMs fail here: they are not calculators, they retrieve the wrong figure, they cannot reliably correct themselves, and their stated confidence is inflated. DecisionGuard is built around five rules, stated as non-negotiable in `CLAUDE.md`:

#	Rule (from `CLAUDE.md`)	How the code enforces it
1	**Code computes, the model narrates.** No number shown to a user may come from an LLM.	Every figure is a `Figure` contract with mandatory `Provenance` (`formula_id`, version, input fact ids, source rows, parameters): `backend/app/contracts/facts.py`. Money is `Decimal`; floats are refused by the `Money` validator (`backend/app/contracts/common.py`). Every number the copilot states is recomputed by the checker before it is shown (`backend/app/agents/checker.py`, §3.8).
2	**Typed facts only.** Engines read the Typed Fact Ledger, never free text.	Engines are pure functions over frozen Pydantic contracts. Fact lookups are exact field-for-field matches; there is no fuzzy or vector lookup (`backend/app/facts/store.py`).
3	**Only anonymised data leaves the server.**	Detectors, token vault, anonymiser and a fail-closed gate (`backend/app/privacy/`). The only door to a model is `backend/app/llm/gateway.py`, which runs the gate on messages and context and refuses the request if any detector fires. The only other optional outbound call is the Sepolia RPC, which carries a 32-byte hash.
4	**Every action has a permission tier** (READ, SUGGEST, DRAFT, EXECUTE, DENY). EXECUTE needs owner approval with a step-up code. Money never moves without a human.	One policy table (`backend/app/security/policy.py`); approvals with fresh TOTP, 24-hour expiry and a re-check at execution (`backend/app/security/approvals.py`). No executor that moves money exists.

#	Rule (from `CLAUDE.md`)	How the code enforces it
5	**Prefer abstaining to guessing.**	Missing eligibility facts give `not_on_record`, never a pass (`engines/financing/eligibility.py`); the cash API answers 422 with what is missing when it has under 8 weeks of history (`api/cash.py`); residual-gap routing abstains when nothing is eligible (`engines/mesh/residual.py`); ratios that cannot be computed honestly are empty with a reason (`engines/analysis/ratios.py`); the copilot abstains on short data, failed checks, low agreement or a noise flip, and logs it (`agents/abstention.py`).

Two further rules shape the data: synthetic data is labelled `data_mode: "synthetic"` everywhere, and when a figure mixes inputs, the most cautious label wins (`combined_data_mode` in `backend/app/facts/provenance.py`).

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2. Architecture

2.1 Components

```
flowchart TB
  user["Owner, accountant, staff or lender viewer<br/>(phone browser, EN / BN / ZH)"]
  subgraph web["Web app: React + TypeScript PWA, self-hosted assets"]
    screens["Screens + provenance drawer + copilot panel"]
    mock["Offline mock API<br/>(development; labelled in the banner)"]
  end
  caddy["Caddy: TLS, security headers,<br/>static web build, /api reverse proxy"]
  subgraph api["FastAPI backend (Python 3.12)"]
    sec["Security layer (WP-11): sessions, TOTP step-up, roles,<br/>permission tiers,
approvals, rate limits, origin check"]
    routers["Routers: auth, approvals, policy, audit, guard, ingest,<br/>facts + explain,
cash, financing, analysis, mesh, copilot, health"]
  end
  subgraph agentsg["Agents (WP-10)"]
    orch["Orchestrator: guard, plan, sub-agents, budget"]
    checker["Checker: recompute every claim"]
    critic["Critic + agreement + self-loops"]
    gw["LLM gateway: privacy gate, fallback chains,<br/>deterministic planner last"]
  end
  subgraph core["Deterministic core (pure engines, Decimal or integer paisa)"]
    ingest["Ingest (WP-02)"]
    facts["Typed Fact Ledger + provenance (WP-01)"]
    cash["Cash engine (WP-04)"]
    fin["Financing engine (WP-06)"]
    ana["Analysis engine (WP-07)"]
    mesh["Mesh engines (WP-08)"]
    chain["Salted SHA-256 commitments + anchorer"]
    priv["Privacy (WP-03)"]
    audit["Hash-chained audit log"]
  end
  db[("PostgreSQL<br/>(SQLite in tests and local dev)")]
  worker["Worker: alert scan 07:30, analysis snapshots,<br/>audit-head anchor 23:55 (Dhaka)"]
  sepolia[("Ethereum Sepolia<br/>32-byte digests only")]
  llm["Hosted LLMs (Gemini, Groq) or local Ollama<br/>(anonymised payloads only)"]
  planned["Receivables WP-05, message drafts WP-12<br/>(reduced scope); eval harness WP-16
runs offline"]

  user --> screens --> caddy --> sec --> routers
```

```
  screens -.-> mock
  routers --> ingest --> facts
  routers --> facts
  routers --> cash
  routers --> fin
  routers --> ana
  routers --> mesh
  routers --> orch
  orch --> cash
  orch --> fin
  orch --> ana
  orch --> mesh
  orch --> checker --> facts
  orch --> critic
  orch --> gw --> priv
  gw -.->|"anonymised text only; skipped offline"| llm
  ana --> cash
  cash --> fin
  cash --> mesh
  mesh --> fin
  cash --> facts
  ana --> facts
  ingest --> priv
  facts --> db
  sec --> audit --> db
  worker --> cash
  worker --> ana
  worker --> audit
  audit --> chain
  mesh --> chain
  chain -.->|"only when CHAIN_MODE is sepolia"| sepolia
  classDef plannedStyle stroke-dasharray: 5 5
  class planned plannedStyle
```

Dashed arrows are optional (a hosted model, Sepolia) or development only (the mock). The dashed box is not in this snapshot.

Component	Path	Role
Contracts	`backend/app/contracts/`	58 frozen Pydantic models, version 1.2.0; unknown fields rejected; exported to `docs/api/contracts.json` and generated into `web/src/api/contracts.ts` (CI fails on drift)
App factory	`backend/app/main.py`	Discovers every `app/api/<name>.py` router and every `app/<area>/app_hooks.py`; CORS is added last so it is the outermost layer; interactive API docs only in development and test
Engines	`backend/app/engines/{cash,financing,analysis,mesh}/`	Pure functions: contracts in, contracts out; no database or network

Component	Path	Role
LLM gateway, agents, checker, critic, agreement, self-loops, abstention, copilot API	WP-10	Built
Agents	`backend/app/agents/`	Orchestrator, tool registry, claims, checker, critic, agreement, self-loops, stakes, budget, abstention
Fact ledger	`backend/app/facts/`	Store, metric registry, provenance graph, masking hook
Ingest	`backend/app/ingest/`	File parsers, normaliser, OCR drafts, preview/commit
Privacy	`backend/app/privacy/`	Detectors, vault, anonymiser, gate, shop-file anonymiser
Security	`backend/app/security/`	Auth, roles, policy, approvals, audit, guard, web hardening
Chain	`backend/app/chain/`	Canonical salted commitments; offline and Sepolia anchorers
Reports	`backend/app/reports/`	Trust Passport, scoped sharing, verification, state adapter (partial)
Worker	`backend/app/worker.py`	Discovers `JOBS` in `app/**/jobs.py`; runs them on APScheduler cron (Asia/Dhaka): alert scan, analysis snapshot freeze, audit-head anchor
Web	`web/src/`	11 screens + public verify page, provenance drawer, EN/BN/ZH, PWA

2.2 Data flow (one request, end to end)

1. **Ingest.** The owner uploads a bKash statement. `POST /api/ingest/preview` detects the format, normalises amounts and dates, clusters counterparties into opaque `PARTY_` tokens, flags duplicates and returns a preview with a keyed digest. Nothing is stored in the ledger yet.
2. **Commit.** `POST /api/ingest/commit` must name the same digest. Only confirmed draft rows (for example OCR lines) are committed. The tenant's ledger revision increases.
3. **Facts.** Ledger metrics (`closing_balance`, `dso`, ...) are computed by registered formulas and stored as `Fact` rows that cite their inputs and source rows. A figure computed against an older ledger revision is never served as current.
4. **Engines.** `GET /api/cash/alerts` runs the forecaster, the Monte Carlo, gap detection, alert tiers and levers. The levers call the financing engine (standby loan, early-pay check) and the mesh engine (netting). Each output number is a `Figure` recorded in the ledger.

5. **Explain.** Any number on screen opens `GET /api/explain/{figure_id}`: formula, inputs with values, and source rows, masked unless the caller holds `source_rows:read` for that tenant.
6. **Act.** An action card (for example "send the application to the lender") is EXECUTE tier. It goes to `POST /api/approvals`; the owner approves with a fresh TOTP code; execution re-checks the policy and the card's SHA-256. Every step is written to the audit chain.
7. **Anchor.** Nightly, the worker commits every tenant's audit-chain head to one digest and anchors it (Sepolia when configured, otherwise offline and labelled offline).

2.3 What the web app talks to

The web calls the API only through `web/src/api/client.ts`. `web/src/api/apiMode.ts` defines three modes: `live` (real backend only; the production default), `mock` (offline fixtures in `web/src/mocks`, always labelled in the banner) and `auto` (development default: real backend first, mock only for routes whose router has not landed).

Live backend in this snapshot	No backend in this snapshot
Sign-in, TOTP, team and lender access, approvals inbox, audit verify	Settings notification preferences (card hidden when the route is absent)
Upload: preview, column mapping, commit (WP-02)	
Cash: forecast, gaps, alerts, what-if (WP-04)	
Financing: match, lender check, application card (WP-06)	
Analysis: statements, ratios, expected vs reality, compare decisions, stability after Compare (WP-07)	
Mesh: obligations, propose, approve with step-up, verify (WP-08)	
Copilot panel and abstention log (WP-10, PR #29)	
Receivables: aging, risk, reminder ladder (WP-05, PR #31)	
Messages: drafts, owner approval with step-up, copy and send log (WP-12, PR #34)	
Reports: Trust Passport versions, sharing consent, public verify page (WP-13, PR #35)	
Market: TCB Dhaka prices, alerts, vendor ranking, negotiation (WP-09, re-included 20 Oct)	
Provenance drawer (`/api/explain`)	

Screens without a backend run only on the mock, which the banner labels. They show how the finished product is meant to work and are not evidence of backend functionality. The demo video uses live screens only (`docs/submission/video-script.md`).

2.4 Deployment

- **Topology** (`infra/docker-compose.yml`): `db` (PostgreSQL 16), `api` (FastAPI), `worker` (same image, `python -m app.worker`), `web` (Caddy serving the built web app and proxying `/api/*` to the API). Secrets come from a server-only env file, never from git.
- **TLS and headers** (`infra/Caddyfile`): automatic Let's Encrypt certificate for `DEPLOY_DOMAIN`; HSTS, a strict CSP (`default-src 'self'`, no third-party origins), `X-Frame-Options: DENY`, `nosniff`,

COOP, compression. The API adds its own headers too, so a request that bypasses Caddy is still covered (`backend/app/security/web.py`).

- **Continuous deployment** (`.github/workflows/deploy.yml`, `infra/remote-deploy.sh`): after CI passes on `main`, the exact commit is shipped, rebuilt, health-checked at `/api/health`, and rolled back to the previous release on failure. It skips until the server secrets exist. Hourly `pg_dump` backups keep 48 copies (`infra/backup.sh`).
- **Location.** The intended host is one server in Singapore (decision D6 in `docs/PLAN.md`: the Gemini API is not served in Hong Kong or mainland China but is in Singapore). No public instance is part of this submission (§7).
- **China reachability by design.** No Google Fonts, CDNs or analytics: fonts and assets are bundled (`docs/packages/WP-14.md`). Swagger UI, which loads from a CDN, is disabled outside development (`backend/app/main.py`). A Playwright test fails if the home page contacts any host other than its own (`web/e2e/smoke.spec.ts`). Measurements from mainland nodes are **pending**.

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3. Core algorithms

3.1 Typed Fact Ledger and provenance (Built)

Files: `backend/app/facts/{store,registry,provenance,masking}.py`.

- **Fact.** `{id, tenant_id, metric, value: Decimal, unit, period, entity_token, provenance, source_refs, data_mode}`. A lookup `(tenant, metric, period, entity)` must match exactly: a metric name differing in case, or a period one day off, is a miss, never a near match.
- **Metric registry.** 17 named, versioned definitions, each with a `formula_id`, a human-readable expression and a pure function: `cash_in`, `cash_out`, `cash_sales`, `credit_sales`, `credit_purchases`, `purchases`, `operating_expenses`, `receivables`, `payables`, `opening_balance`, `net_sales`, `net_cash_flow`, `closing_balance`, `gross_profit`, `gross_margin`, `dso`, `dpo`. A released definition never changes; a new rule is a new version, so old figures still explain themselves with the old expression. DSO and DPO are on a credit basis (`dso = receivables / credit_sales x days_in_period`) because cash sales never wait.
- **Provenance graph.** `compute()` stores each result with the closure of inputs underneath it, cited as `txn:<id>`, `inv:<id>`, `bill:<id>` or `fact:<id>`. Engines outside WP-01 record their own figures with `record_computed()` / `record_figure()`. The cash API records every figure it returns, with the `closing_balance` fact as its cited input (`backend/app/api/cash.py`, class `Recorder`).
- **Staleness.** Every change to a tenant's ledger bumps a revision. Computed facts record the revision they were confirmed against, so a figure that may predate newer rows is recomputed (or reported stale), never served as current.
- **Masking.** Source rows pass through one masking hook for every caller. The installed policy (`backend/app/security/roles.py`, `policy_masker`) shows rows only to a same-tenant caller holding `source_rows:read`; others get keyed, non-reversible references.

3.2 Ingestion (Built)

Files: `backend/app/ingest/`.

- **Formats:** bKash and Nagad statements, generic bank CSV, ledger CSV/XLSX with a column mapping step, ledger photos. Detection matches header names (English, Bangla, Banglish) exactly after normalisation; preamble lines above the table are skipped (`formats.py`).

- **Normaliser** (`normalise.py`): Bangla digits, Tk/৳/BDT/টাকা markers, lakh grouping and লাখ/কোটি/হাজার words, month names in three scripts, Bengali calendar dates, day/month order decided per file by evidence. A value that cannot be read with certainty raises an error with a reason key and becomes an unparsed row or a draft, never a guess.
- **OCR** (`ocr.py`): local Tesseract with `ben+eng`; the photo never leaves the server. Each line becomes a draft with a confidence; low-confidence drafts are flagged and never committed without the owner ticking them.
- **Preview/commit** (`preview.py`): the preview stores an HMAC-SHA256 of the uploaded bytes (key derived from the server secret, bound to the tenant). The commit must present the same digest. Duplicates are checked at preview and again at commit. Uploads are capped at 10 MB; previews expire after 24 hours.

3.3 Cash engine: forecast, Monte Carlo, gaps, alerts, levers (Built)

Files: `backend/app/engines/cash/`. API: `backend/app/api/cash.py`. Feature spec: FEATURES Q1, Q3.

Baseline forecast (`baseline.py`, `sales.py`, `delays.py`). For each of the next 91 days:

```
balance(day) = opening balance + expected inflows - expected outflows

sales(day)        = level x weekday factor x month-end factor x festival factor
collections       = sum over open invoices of owed x P(customer pays on that day)
new credit sales  = recent credit-sales rate x share of invoices paid within n days of issue
outflows          = recurring obligations on their schedule
                  + open bills on the day this business usually pays that supplier
                  + new bills, cash purchases and running costs at their recent daily rate
```

All factors are estimated from the shop's own history up to the forecast date only: weekday and month-end factors from the last 26 weeks, the level from the last 8 weeks, festival factors (Eid-ul-Fitr, Eid-ul-Adha, Pohela Boishakh; `data/public/calendar/bd_festivals.yaml`) from earlier occurrences, with a run-up of 10 days and 3 days after. A factor without evidence stays at 1. Payment delays are each customer's own `paid date - due date` history (the tenant-wide distribution when a customer has fewer than 3 settled invoices), conditioned on the invoice still being unpaid today. If no observed delay is left, the invoice contributes nothing: the forecast prefers under-counting cash to inventing it. Disputed invoices are not expected.

Why not the ML delay model: on a time split of the synthetic data, LightGBM scored MAE 2.92 days against 2.83 for the per-customer history, and the decision rule required a 5% improvement (source: `ml/artifacts/payment_delay_metrics.json`).

Monte Carlo (`simulate.py`). 1,000 paths, seed 20261020, NumPy PCG64, amounts in integer paisa and ratios in integer basis points, so every band is exact and reproducible:

- sales noise: the shop's own observed/expected ratios on past plain days, resampled in 28-day circular blocks (a slow fortnight stays slow);
- collections: one lateness draw per customer per path, applied to all of that customer's invoices (a slow payer is slow everywhere);
- costs and new bills: past days' actual spending replayed in 28-day blocks, so one-off shocks recur as often and as large as they did;
- obligations and open bills: certain.

Percentiles are nearest-rank order statistics (P10 = the 100th lowest of 1,000), so `P10 <= P50 <= P90` holds by construction. A what-if or a lever replays the **same draws** (common random numbers): the difference between two runs is the change, never luck.

Buffer target (`buffer.py`): the 95th percentile, over paths, of the deepest dip below the likely (P50) line. Keeping P50 at or above it keeps the shop out of the red in 19 of 20 paths.

Gaps (`gaps.py`): a *risk day* has more than 5% of paths below the buffer; a *gap* is a run of risk days in which more than 20% of paths dip. For each gap the engine reports the first breach, the worst day, the probability (paths below the buffer on at least one gap day), the expected shortfall `mean over paths of max(0, buffer - lowest balance in the gap)`, the lead time and the three largest drivers in Tk (bills, obligations, late customers, sales below norm). Five leading indicators fire from the ledger before any crossing: DSO rising, a top customer slowing, stock building, sales below the seasonal norm, payroll or EMI at risk.

Alert tiers (`alerts.py`), exactly as FEATURES Q1. "P within a-b" = share of paths below the buffer on at least one gap day between day a and day b:

Tier	Trigger (any one)	Delivery
Critical	P within 1-7 > 20%; P within 1-21 > 60%; a salary or EMI day in the gap with P50 < 0	in-app + email, pinned on Today
Warning	P within 8-21 > 20%; P within 22-45 > 50%	in-app + email
Watch	P within 22-90 > 20%	in-app + weekly digest

Alert fatigue: a repeat is sent only if it escalates, or the forecast moved by more than 3 days or 10% of the shortfall, and never inside the cooldown (24 h Critical, 3 days Warning, 7 days Watch). Email delivery belongs to WP-12 and is not built; the daily scan job records deliveries.

Levers (`levers.py`). Each lever is a scenario replayed on the gap's own draws; its effect is `shortfall without - shortfall with` on the same paths. Only levers that reduce the gap are offered, ranked cheapest first: re-time a payable within terms, stagger a purchase week, ask for an advance, offer a 2% early-pay discount (only if its annualised cost is below the cheapest eligible loan), net circular debt through the mesh, or arrange a standby loan (WP-06 ranking, sized to cover the gap in 19 of 20 paths). Each becomes an action card with its policy tier.

Abstention. With fewer than 8 weeks of transactions, or no opening balance on record, the cash API answers 422 with an `ApiError` naming what is missing (`api/cash.py`, `MIN_HISTORY_DAYS = 56`).

Worked example (synthetic grocery shop `sme_grocery`, as of 2026-09-30; computed for this document with the repo's engines, see Appendix A):

Quantity	Value
Opening balance	Tk 427,842.00
Buffer target (P95 dip)	Tk 296,481.07
Band on 2026-12-30 (day 91): P10 / P50 / P90	Tk 379,696.79 / 511,060.52 / 638,980.00
Plain forecast	no gap; runway = full 91-day horizon
What-if: sales -10%, every customer pays 7 days later (same draws)	4 gaps

Quantity	Value
First gap: days 4-13, first breach day 10	P = 0.2720, expected shortfall Tk 11,901.90, tier **Warning** (P within 8-21 > 20%)
Its top drivers	two supplier bills of Tk 82,618.00 each, an obligation of Tk 35,000.00
Best lever: stagger a purchase week	cost Tk 0, shortfall reduced by Tk 7,602.01, P after = 0.0960, EXECUTE card
Standby loan lever	cost Tk 2,697.69, shortfall reduced by Tk 4,455.14, DRAFT card
Last gap: days 29-91	P = 0.9770, expected shortfall Tk 159,197.94, tier Warning (P within 22-45 > 50%)

Backtest (`backtest.py`, FEATURES N15). Walk-forward: at each origin (every 14 days) the forecaster sees only what was known that day, forecasts 91 days with 1,000 paths, and is scored against what happened. Results on the **synthetic** shops (source: `docs/packages/WP-04.md`; re-run for this document with `python -m app.engines.cash.backtest --sme <id>`, identical output; the last two columns are from that re-run):

SME (synthetic)	MAPE wk 1 ours / naive	MAPE 13 wks ours / naive	P10-P90 coverage	gap episodes	hit rate	median lead	alerting gaps / did not happen
grocery	9.6% / 15.6%	22.4% / 20.7%	77%	5	100%	76 d	50 / 34
pharmacy	7.9% / 15.5%	16.3% / 17.9%	81%	7	100%	75 d	44 / 27
fabric	9.7% / 21.6%	33.7% / 31.0%	73%	24	100%	84 d	38 / 10
garment	23.2% / 41.2%	30.5% / 51.8%	61%	10	90%	81 d	28 / 0
hardware	7.9% / 18.7%	15.4% / 24.0%	83%	8	100%	87 d	52 / 32

Reading: the forecast beats "balance stays the same" in week 1 for every shop, which is the alert window. It loses over 13 weeks at the grocery and fabric shops. Alerts are over-cautious at three shops: many alerting gaps did not happen. These are synthetic shops generated by our own code, so this checks the harness, not real-world accuracy.

Real-data backtest (`ml/notebooks/01_forecast_backtest.py`, run 20 Oct; results in `ml/artifacts/forecast_backtest.json` and `forecast_backtest_origins.csv`). Data: UCI Online Retail II (CC BY 4.0), a real UK online retailer's 1,067,371 invoice lines turned into 739 days of net cash-in (Dec 2009 to Dec 2011). 19 walk-forward origins, 14 days apart, each forecasting 28 days from data before the origin only. Cash gaps are judged on a simple outflow rule (weekly supplier run at 95% of average inflow, rent on the 1st) with the opening balance one day of sales above the expected low point.

Model	WAPE	Weekly MAPE	80% band covers	Real gaps flagged	False alarms	Lead time
Seasonal naive (same weekday last week)	43.4%	24.8%	71.2%	3 of 5	42.9%	10.7 d
ETS (the method family our engine uses)	36.3%	22.9%	71.8%	3 of 5	28.6%	10.7 d

Model	WAPE	Weekly MAPE	80% band covers	Real gaps flagged	False alarms	Lead time
LightGBM quantile	34.1%	22.2%	54.0%	3 of 5	50.0%	10.7 d

Reading: on real data we did not generate, ETS beats the naive baseline on every error measure and raises fewer false alarms; LightGBM is slightly more accurate but its 80% band covers only 54% of days, so its warnings are overconfident, and ETS is the model the notebook chooses. Gaps that were flagged were flagged about 11 days ahead. Only 5 real gaps occurred, so the hit and false-alarm rates are rough; this is one retailer, not a Bangladeshi shop.

Real Bangladeshi data (same notebook with `HISHAB_DAILY_PATH`, run 20 Oct; results in `ml/artifacts/forecast_backtest_bd.json`). Data: a Bangladeshi retailer's daily units sold for one product, 1 Jan 2013 to 31 Dec 2017 (Mendeley 10.17632/xwmbk7n3c8.1, CC BY 4.0, copy in `data/public/retail_bd/`), used as a stand-in for daily inflow. 103 origins from 2014 on, the same 28-day horizon and gap rule.

Model	WAPE	Weekly MAPE	80% band covers	Real gaps flagged	False alarms	Lead time
Seasonal naive	26.1%	11.1%	77.7%	41 of 45 (91%)	15.5%	17.1 d
ETS (chosen)	19.9%	10.8%	73.0%	37 of 45 (82%)	17.2%	17.8 d
LightGBM quantile	19.9%	9.1%	56.9%	39 of 45 (87%)	15.5%	17.5 d

Reading: on real Bangladeshi sales, ETS cuts the daily error by about a quarter against the naive baseline, but it does not flag more gaps: the naive forecast caught 4 more of the 45, at a similar false-alarm rate. Better accuracy did not mean better warnings here, so the alert layer, not only the forecaster, needs tuning on real shop data. One product of one shop, units not taka.

3.4 Financing: true cost, eligibility, ranking, lender check (Built)

Files: `backend/app/engines/financing/`. API: `backend/app/api/financing.py`.

True cost (`cost.py`). Every option becomes dated cash flows from the borrower's side:

```
day 0      + principal - upfront fees - compulsory deposit
later      - instalments, periodic fees, maturity fees; + deposit returned at maturity

EAR = the r that solves   sum_i  cf_i / (1 + r)^(day_i / 365)  =  0
```

The rate is found by bisection in `Decimal` at 34 digits of precision (bracket doubled until the present value turns positive, then up to 200 halvings to a width of 1e-12). Bisection is slower than Newton's method but always converges and is deterministic. Rate conventions as Bangladeshi lenders publish them: reducing balance (EMI), flat (interest on the original principal), daily debit (flat interest repaid in equal daily debits, as nano loans do), discount (interest taken up front) and none (owner top-up, netting). The early-pay discount used by levers is annualised as `(1 / (1 - d))^(365 / days_early) - 1`.

Worked example: Tk 50,000 for 180 days (computed with `true_cost()` on rows of `data/public/financing/bd.yaml`):

Product (catalogue id)	Published rate (basis)	Structure, fees	EAR	Total cost
Bangladesh Bank CMSME WC refinance (`bd-bb-cmsme-wc-refinance-2026`)	9.00% (regulatory cap)	reducing, 0.5% upfront	11.29%	Tk 1,570.67
bKash digital loan, City Bank (`bd-citybank-bkash-loan`)	9.00% (last published)	daily debit, 0.575% upfront	22.15%	Tk 2,506.68
Prime Chalti, Prime Bank (`bd-primebank-chalti`)	15.00% (rate-sheet upper)	reducing, 0.5% upfront	18.11%	Tk 2,460.14
PKSF microenterprise (`bd-pksf-microenterprise`)	24.00% (MRA cap)	reducing, no fee	26.82%	Tk 3,557.74

The same 9% headline rate costs 11.29% a year as a reducing-balance loan and 22.15% as a daily-debit loan, because daily repayment halves the average balance while interest is charged on the full principal. Likewise a "2% off if paid 10 days early" discount costs 109.05% a year, and 2% for 30 days costs 27.86% (`early_pay_discount_cost`), which is why levers offer a discount only when it is cheaper than borrowing.

Eligibility (`eligibility.py`). Each published rule (for example `business_age_months >= 36`, `trade_licence exists`) returns `pass`, `fail` or **`not_on_record`** when the fact is missing. A missing fact never passes. Amount, tenor and purpose limits can only fail. Overall status: any fail is fail; else any missing is `not_on_record`; else pass. The engine also lists exactly which facts the owner could add (`missing_facts`). In the synthetic profiles, `bkash_account_active`, `no_default_record`, `owner_is_woman`, `enterprise_closed` and `collateral_available` are not on record, so products that need them come back `not_on_record` (`docs/packages/WP-01.md`, Questions).

Repayment fit (`fit.py`). Given WP-04's band, each day's balance is approximated as normal with mean P50 and spread `(P90 - P10) / 2.5631`; the option's cumulative cash flows are added, and the highest daily probability of falling below the buffer is reported. This is a lower bound on the chance of any breach and is labelled an approximation.

Ranking (`ranking.py`). Score 0-100:

```
score = 0.45 cost + 0.25 fit + 0.15 approval + 0.10 speed + 0.05 lender
cost     = max(0, 100 - 2.5 x EAR%)          (4% scores 90; 40% or more scores 0)
fit      = 100 x (1 - P(balance < buffer))
approval = 100 x heuristic by product kind, x 0.6 if facts are missing, x 0.5 if collateral
is needed and not on record
speed    = max(0, 100 - 3 x typical days to cash)
lender   = lender score (60 with no track record)
```

Ineligible options stay in the list with their reasons but score 0 and always rank below every option that might work. Ties break on lower EAR, then product id, so the order is deterministic. The approval and speed tables are documented assumptions shown to users, not lender data.

Lender check (`lenders.py`, FEATURES N20). From verified track-record events:

```
score = 100 - 50 x (hidden-fee incidents / n) - 30 x (disputes / n)
            - 4 x max(0, mean(realised EAR - quoted EAR)) - 15 if licence unknown
score = 0 if the licence is not found or revoked;   score = 60 with no events (cold start)
```

Application card (`application.py`). The best passing option becomes a `submit_loan_application` card: EXECUTE tier, owner approval with step-up, 24-hour expiry. Its payload holds only facts we have, the facts still missing, the EAR, the rate basis and the source URL. Drafting (`draft_loan_application`) is DRAFT tier.

What the API does not do yet (stated in `backend/app/api/financing.py`): the WP-04 band is not passed in, so every option carries `fit.no_forecast`; no lender has a stored track record; no licence registry is loaded, so licence status is `unknown`. The non-loan options in `nonloan.py` (owner top-up, supplier term extension, buyer early pay, mesh netting) are priced by the same engine but reach users through the cash levers, not through `/financing/match`.

3.5 Liquidity mesh: netting, consent, settlement, routing, attestations (Built; API in PR #26)

Files: `backend/app/engines/mesh/`; API: `backend/app/api/mesh.py` (PR #26). Feature spec: FEATURES N14, Q4.

Graph (`graph.py`). Only open BDT obligations between two opted-in tenants enter a directed graph, debtor to creditor, amounts in integer paisa. Groups with circular debt are the strongly connected components of size 2 or more.

Netting by minimum-cost flow (`netting.py`). For each group:

```
node demand(t) = sum owed to t - sum t owes          (net position; > 0 receives)
edge d -> c    = capacity: what d owes c;  cost 1 per paisa
minimise total cash moved, subject to every node meeting its demand
(networkx.min_cost_flow)
```

The original debts are themselves a feasible flow, so a solution always exists; the optimum is the original minus the largest circulation that can be cancelled. Settlements run only along existing debts, so nobody pays a firm it never owed. An opt-in mode (`allow_new_links`) lets any net debtor pay any net creditor and reaches the theoretical minimum, the sum of positive net positions. Invariants, property-tested with Hypothesis on random graphs in `backend/tests/mesh/test_netting.py`: every net position is unchanged; along debts, no edge carries more than was owed; with new links, nobody pays more than their total debt; net total never exceeds gross; no payment cycle is left.

Worked example: the synthetic demo ring (`data/synthetic/universe/mesh_obligations.csv`; computed with `build_graph` and `propose_rounds`):

Obligation	Debtor -> creditor	Amount
RING-FG-001	garment -> fabric	Tk 450,000.00
RING-GH-001	hardware -> garment	Tk 320,000.00
RING-HF-001	fabric -> hardware	Tk 380,000.00
Gross circular debt		**Tk 1,150,000.00**
Net positions	fabric +70,000; hardware +60,000; garment -130,000	
Netting along debts	garment pays fabric 130,000; fabric pays hardware 60,000	**Tk 190,000.00 cash**
Netting with new links	garment pays fabric 70,000 and hardware 60,000	**Tk 130,000.00 cash**

Every net position is the same after either settlement (checked in the run).

Consent (`consent.py`). Approving a round is EXECUTE tier: every participant must approve with a verified step-up code; one rejection voids the round; a repeated approval is a no-op; rounds expire after 48 hours.

Settlement (`settlement.py`). Only an approved, unexpired round whose obligations are all still open settles. Each participant gets a receipt: what it pays and receives, which obligations were discharged, and `cash saved = debts cleared - cash paid`. The settled record is then sealed and anchored (§3.6). Moving money between SMEs can count as lending or deposit-taking, so real funding is meant to route through licensed lenders (`docs/CONTEXT.md` §3); no money moves in this snapshot.

API (`backend/app/api/mesh.py`, PR #26). `GET /api/mesh/obligations` (the caller's debts and those inside its netting group), `POST /api/mesh/rounds/propose` (DRAFT tier: min-cost netting of the caller's group, along existing debts), `POST /api/mesh/rounds/{id}/approve` (EXECUTE tier: owner with `X-Step-Up-Code`; the last approval settles the round and anchors its salted digest), `POST /api/mesh/rounds/{id}/reject`, and `GET /api/mesh/rounds/{id}/verify`. The salt never leaves the server. Propose, approve, settle and reject are written to the audit chain. The only obligations in this snapshot are the synthetic demo ring; there is no opt-in flow for real firms.

Residual-gap routing (`residual.py`). After netting, a participant may still owe more cash than it has: `gap = max(0, cash_out - cash on hand)`. Cash on hand comes from the fact ledger; a tenant without that fact is treated as having none (fail closed). The gap is priced with the WP-06 engine for the exact amount and bridge tenor; ineligible products are dropped and the lowest **total taka cost** wins (ties: EAR, then WP-06 rank). If the cheapest is not WP-06's top balanced rank, it is flagged `cheaper_than_top_ranked`. With nothing eligible, the router abstains (`mesh.residual.no_eligible_option`) and the round cannot settle for that participant until the owner decides.

Counterparty attestations (`attestations.py`, FEATURES Q4). Debtor and creditor each sign the SHA-256 commitment of the same obligation record (who owes whom, how much, by when). The record is **peer-verified** only if both parties confirmed that exact digest and nobody filed a dispute. Attestation digests use a fixed public salt so either side can recompute them.

Lender events (`lender_events.py`). A loan becomes a track-record event for the lender check only when it is fully repaid and every repayment obligation is peer-verified. Then `realised APR` = the EAR (same bisection, §3.4) of the cash actually received and repaid; `hidden fee` = the borrower received more than Tk 1 less than principal minus disclosed fees; a dispute counts against the lender without using the contested amounts. Unverified loans are skipped with a reason: no record is better than an unverified one.

3.6 Tamper evidence: salted commitments and Sepolia anchoring (Partial)

Files: `backend/app/chain/{digest,anchor}.py`.

```
digest = SHA-256( "hishab:v1:" + kind + ":" + salt + ":" + canonical JSON )
canonical JSON = sorted keys, no spaces, decimals as strings;   salt = 32 random bytes (hex)
```

The salt stays off-chain with the record. Without it, nobody can brute-force small amounts or party names from a public hash. Only the 32-byte digest goes on chain: the Sepolia anchorer sends a zero-value self-transfer whose data field is exactly the digest (chain id 11155111), and `payload_for()` refuses anything that is not a 64-character lowercase hex digest. Verification recomputes the digest from the record and salt, reads the transaction's data field, and reports which of four checks failed (not anchored, not found on chain, receipt differs from chain, record changed after anchoring). The same mechanism seals netting rounds, Trust Passport versions (`backend/app/reports/passport.py`) and the daily audit-chain heads.

What runs today: the daily job `security.anchor_audit_heads` (23:55 Dhaka) anchors through `SepoliaAnchorer` when `CHAIN_MODE=sepolia` and an RPC URL and key are configured, otherwise through `OfflineAnchorer`, labelled `offline`. Settled netting rounds are anchored the same way by the mesh API (PR #26), and Trust Passport versions when they are issued (`/api/reports/passport`, PR #35). Tests use a fake web3 object and never touch a node (`backend/tests/mesh/test_sepolia.py`). **No Sepolia transaction hash is recorded in this snapshot.**

3.7 Privacy pipeline (Built; on every model call)

Files: `backend/app/privacy/`. Feature spec: FEATURES N6, N18, P11.

1. **Detect** (`detectors.py`): Bangladesh mobile numbers (bKash and Nagad accounts are mobile numbers), NID, payment cards (Luhn), long account numbers, emails, Dhaka addresses and area names, person and shop names from the party registry and a name list. Bangla digits are recognised everywhere. Fail closed: any standalone run of 10 or more digits that is not a phone number is treated as sensitive.
2. **Tokenise** (`vault.py`): `PERSON_xxxxxxxxxx = HMAC-SHA256(tenant | kind | normalised value)`, truncated, so tokens are stable inside a tenant and different across tenants. Originals are stored with AES-256-GCM (random 96-bit nonce; tenant and token as associated data, so a ciphertext copied elsewhere fails to decrypt). HMAC and encryption keys are derived separately from `VAULT_KEY` with HKDF. Restoring an original needs an authoriser; the default refuses everyone.
3. **Anonymise** (`anonymiser.py`): tokens become neutral labels in order of appearance ("Customer 1", "Firm A"); dates become relative ("month t-2"); place names and rare product terms become `[place]` / `[item]` (a memorisation guard); optionally amounts are multiplied by a secret factor so ratios survive and scale does not (P11). `restore()` maps labels back to tokens. Numbers in a model's answer are never restored or trusted.
4. **Gate** (`gate.py`): `assert_safe(payload)` runs every detector on the serialised payload and raises if any fires. The error names only kinds and counts, never values.

In WP-10 the orchestrator anonymises the owner's question before the planner sees it, and the narrator and critic receive only verified claim texts (figures, product names, party labels, no personal data; amounts are not scaled for these calls). `prepare_messages()` offers one label set across several messages for callers that need it. The gateway (`backend/app/llm/gateway.py`) runs the gate on the messages *and* the structured context before any provider is tried. If a detector fires, the request is refused and counted, never sent. Model answers come back with labels, which are mapped to tokens; numbers in them are never used (§3.8). For field data (WP-20) the shop-file anonymiser (`python -m app.privacy.shopfiles`) tokenises CSV/XLSX columns, refuses any file with leftovers, and writes a counts-only report.

3.8 Analysis: statements, ratios, expected vs reality, scenarios, stability (Built)

Files: `backend/app/engines/analysis/`. API: `backend/app/api/analysis.py`. Feature spec: FEATURES N1, N4, N10, N12, N16.

Statements (`statements.py`). Cash basis: income is money received, costs are money paid. Every transaction falls in exactly one line, so the statements reconcile by construction:

```
net_profit + loan flows + owner flows + transfers = cash_in - cash_out = net cash flow
closing cash = opening cash + net cash flow;     total assets = total liabilities + net
position
```

Stock and loans appear only when a stock count or loan statement is on record; otherwise they are left out and named in `missing`.

Ratios (`ratios.py`). 13 ratios, each a figure over statement lines with provenance: gross and net margin, current and quick ratio, DSCR, interest cover, DSO, DPO, DIO, cash conversion cycle, burn, runway and top-customer share. A ratio that cannot be computed honestly (a fact not on record, a zero denominator, no debt) is empty with a reason key such as `ratio.missing.stock_value`. Interpretations come from named thresholds, never from a model.

Expected vs reality (`variance.py`, `store.py`). Each week and quarter, WP-04's baseline, fitted only on rows dated up to that day, is frozen as an immutable snapshot (a second write is refused). When the period ends, seven cash-flow lines and two totals are compared with what happened, ranked by the size of the gap. Accuracy = 1 - weighted absolute percentage error. Recent checks become a sales and a cost correction (credibility-weighted towards 1 while there are few), handed to WP-04 as a what-if. The nightly job `analysis.snapshot_freeze` takes the snapshots.

Scenarios (`scenarios.py`). WP-04's 1,000 seeded paths are split into 40 macro states drawn by Latin hypercube sampling: a sales level shift (sd 8%), input prices (sd 5%), customers paying 0, 7 or 14 days later (70/20/10% of states), rates up to +2 points, and a one-off shock in 10% of states. Six decisions (do nothing, take the best eligible WP-06 loan, offer an early-pay discount, re-time a supplier, delay a purchase, net through the mesh) are replayed on the same paths and states. For each: expected outcome, mean of the worst 5% of outcomes, and P(buffer breach). The pick: among decisions within 5 points of the lowest breach risk, the highest (expected + worst 5%) / 2; ties go to the cheaper decision.

Stability score (`stability.py`). A recommendation is re-made under 10 variations: 6 economic stresses, where the advice may change (the largest debtor pays 30 days late, sales -10% and -20%, inputs +8%, rates +2 points, a surprise bill of a month's fixed costs due in 7 days), and 4 noise variations, where it must not (rows shuffled, parties renamed, question reworded, dates written another way). Score = share of variations in which the choice held; `what_would_flip` names the stresses that changed it; any noise flip marks the advice unreliable. The noise check found a real bug: WP-04 broke ties between equally owed customers on their token, so renaming two customers swapped their lateness draws. WP-04 now breaks such ties on record ids (`tests/cash/test_names.py`).

Worked example (synthetic hardware shop `sme_hardware`, quarter Jul-Sep 2026; computed for this document through `/api/analysis/*` on the synthetic universe, see Appendix A):

Statement line	Value
Revenue (cash sales Tk 33,81,248 + collections Tk 14,70,590)	Tk 48,51,838.00
Purchases	Tk 37,81,353.00
Gross profit	Tk 10,70,485.00
Operating expenses (rent, salaries, utilities)	Tk 4,93,602.00
Net profit (cash basis)	Tk 5,76,883.00
Net cash flow = 5,76,883 - loan repayments 1,26,000 - owner drawings 3,18,599	Tk 1,32,284.00
Closing cash = opening Tk 7,96,734 + 1,32,284	Tk 9,29,018.00 (equals the cash forecast's opening balance)
Named as missing	stock value, loan outstanding

Ratios: gross margin 22.1%, net margin 11.9%, quick ratio 0.91 ("short of current cover"), DSCR 4.58 ("comfortable"), DSO 40.7 days, DPO 33.6 days, top-customer share 4.2%. Current ratio, DIO and the cash conversion cycle are empty (`ratio.missing.stock_value`); interest cover is empty (`ratio.missing.interest_expense`).

Compare decisions (buffer = WP-04's target Tk 9,19,664.57; 1,000 paths, seed 20261020):

Decision	Expected outcome	Worst 5%	P(buffer breach)
Do nothing	Tk 16,19,493.02	Tk 4,62,756.51	0.9390
Take the best eligible loan (paid out 30 Oct)	Tk 16,16,268.61	Tk 4,59,455.33	0.9240
Offer 2% off the open invoice with the most cash still expected (pay by 2 Oct)	**Tk 16,11,893.02**	**Tk 4,55,156.51**	**0.5190**
Delay a purchase week	Tk 16,19,493.02	Tk 4,62,756.51	0.9380
Net through the mesh	Tk 16,19,493.02	Tk 4,62,756.51	0.8310
Re-time a supplier	skipped: the shop pays no open bill before its due date		

The discount costs exactly Tk 7,600 (2% of a Tk 3,80,000 invoice) and cuts the breach risk from 93.9% to 51.9%. The loan barely helps because it can only pay out on 30 Oct, after the gap starts. Stability: 0.80 (8 of 10 held); the choice flips under sales -20% and under a surprise bill; all four noise variations held, so the advice is not marked unreliable.

3.9 Agents and the LLM layer (Built on branch `wp-10-agents`; PR pending)

Files: `backend/app/llm/`, `backend/app/agents/`. API: `backend/app/api/copilot.py` (`POST /api/copilot/ask`, `GET /api/copilot/today`, `GET /api/copilot/abstentions`). Feature spec: FEATURES N8-N11, N17, P9, P13.

Gateway (`llm/gateway.py`, `llm/providers.py`). One OpenAI-compatible client; only the base URL, key and model change. Slots: the brain (Gemini Flash), voters (gpt-oss-120b and Qwen, falling back to gpt-oss-20b and Gemma on Groq), the critic (the same pool, never the brain's model family) and a private slot (Ollama on our own machine). Each request walks its chain: cache, key present, within our own per-slot quota (requests per minute and per day, with doubling cooldowns), send, valid JSON. Any failure moves to the next link. **The last link of every chain is a deterministic planner, so the app works with no model at all.** The cache is keyed by model and the hash of the anonymised prompt. Tests replay recorded fixtures (`LLM_MODE=fixtures`).

One question, end to end (`agents/orchestrator.py`):

1. **Guard.** The question passes WP-11's injection guard before any model sees it; a hit is quarantined and only its hash is audited.
2. **Plan.** The brain proposes intents and tools from the anonymised question. Code repairs the plan (`planning.py`): unknown intents and tools are dropped, dependencies run first, and what-if shocks are read from the owner's words by rules, never by a model.
3. **Dispatch.** One sub-agent per intent, in parallel threads, sharing a memoised tool runner. A `Budget` caps tool calls, model calls and tokens per run (P13). Tools (`tools.py`, `toolset.py`) have validated parameters and a minimum history, are never EXECUTE tier, and every figure they return must cite facts, rows or parameters: `cash.forecast`, `cash.gaps`, `analysis.scenarios` (WP-04 what-if), `financing.match` (sized to the gap, with the forecast band), `analysis.ratios`, `mesh.propose`.
4. **Compose.** Every number becomes a `Claim{text, value, formula_id, inputs}` rendered by code in English, Bangla or Chinese. The brain only narrates the claim texts. At most three action cards the caller's role may act on are attached (P9).
5. **Check** (`checker.py`). Each claim must match its figure and state its value as code renders it, and is then recomputed independently:

Source	Recomputed from
WP-01 metrics	the cited input facts re-read from the ledger, through the registry *and* through SymPy on the published expression, so the code and the formula a user reads must agree
WP-04 cash figures	a freshly drawn simulation with the figure's seed, buffer and scenario, through the engine's own figure and lever functions
WP-06 costs	the catalogue product's cash flows for the figure's principal and tenor
WP-08 netting	the netting round, recomputed from the cited obligations

A mismatch is rejected; a claim with no recomputation path is unverifiable. Either way it is removed from the answer and logged. A narration containing a number that no verified claim states is replaced by the claim texts. The tests inject a wrong number into each of the 26 kinds of stated number. 6. **Stakes** (`stakes.py`): low = information; medium = drafts, netting, what-ifs; high = borrowing or moving money. Stakes decide how much checking happens before the owner is asked. 7. **Critic** (`critic.py`, medium stakes and up). A model from a family that took no part in the answer argues the case against, given the verified claims and the risks code found (approval slower than the gap, unknown eligibility, still below the buffer, netting needs every firm's consent, short history, synthetic data). An objection that states a number no verified claim states, or no objection at all, falls back to the worst risk code found. 8. **Agreement** (`agreement.py`, medium and up). The judgement the answer rests on goes to two other models, next to code's own ranking as the first

vote. Agreement rate = share of votes for the winner; the threshold is a majority, 0.66 or everyone, by stakes. The deterministic rules never vote, so offline or out of quota **no agreement is claimed**. 9. **Self-loop** (`selfloop.py`). The option the answer rests on is re-made under stress and noise (no variations at low stakes, 2 at medium, 4 at high) and reported with what would flip it. 10. **Abstain** (`abstention.py`). The answer is withheld when no claim recomputed, when any claim was rejected at high stakes, when the voters do not back code's choice at the stakes' threshold, or when the decision flips on noise. Every abstention names what is missing, is stored, and is appended to the audit chain as `copilot.abstain`.

Worked example (synthetic hardware shop; offline: no model was called, so the deterministic planner planned and narrated, the critic objection is the worst risk found by code, and no agreement is claimed; computed for this document with `POST /api/copilot/ask` on branch `wp-10-agents`, see Appendix A):

Question	Tools run	Claims stated / verified	Action cards	Critic	Stability
"Will I run out of cash in the next month?"	`cash.forecast`, `cash.gaps`	12 / 12	2 x ask a buyer for an advance (EXECUTE)	"Asking a customer to pay early can strain the relationship and may be refused."	0.75: if customers pay 7 days later, netting becomes the better first fix
"How big a loan can I safely take?"	`cash.gaps`, `financing.match`	14 / 14	2 x advance (EXECUTE), 1 x propose a netting round (DRAFT)	"We could not confirm that you qualify for Startup financing ...: the lender will ask for facts DecisionGuard does not hold."	1.00

From the second answer, every figure checked: cash gap starting in 5 days with probability 86.70% and expected shortfall Tk 73,460.90; the cheapest fix closes Tk 71,736.88 of it at no cost; the Bangladesh Bank startup fund costs 4.07% a year, Tk 490.28 in total for this need, and leaves a 47.70% chance of still falling below the buffer.

No live-model trace is part of this document. Live answers can be recorded as fixtures and JSONL traces with `python -m app.llm.live_check` (keys required); the evidence index lists any that were.

3.10 Receivables, messages, evaluation, and what was cut

- **Receivables** (WP-05, reduced): aging buckets, an empirical 0-100 risk score per open invoice with its top 3 factors (monotone in lateness, property-tested), and a reminder ladder (-3, 0, +3, +10, +20 days) giving one EXECUTE `send_reminder` card per customer (`backend/app/engines/receivables/`, `GET /api/receivables`, `GET /api/receivables/plan`).
- **Message drafts** (WP-12, reduced): English and Bangla templates filled from typed values only (an unfilled variable is refused), one draft per reminder, owner approval and a send log (`backend/app/messaging/`, `/api/messages/*`). There is no sending channel: the owner copies the approved text into their own chat app.
- **Evaluation harness** (WP-16): `npm run eval` runs the tasks offline and writes per-step JSONL logs to the evidence bundle (`backend/eval/`, results in `docs/submission/evidence/`). On the

submission commit: 62 of 62 tasks pass (functional 26/26, adversarial 36/36). Read the adversarial number with care. The rule guard first caught only 3 of 12 new, unseen attacks (11 of 12 were still contained, and approval held in all 12). We then widened each attack family's rules to intent level, with no new false alarms on the 96 benign corpus texts, and the 12 are now caught. Because we saw them while doing so, 12/12 is a regression result, not proof of generalisation. On the final audit's fresh held-out set (24 attacks and 24 innocent look-alikes in four languages, never seen while widening) the rules caught 15 of 24 attacks and wrongly quarantined 10 of 24 innocent questions (`docs/handover/fresh_injection_result.txt`).

- **Re-included 20 Oct (reduced):** market watch and negotiation (WP-09: N3, N19) on TCB's published Dhaka price lists; vendor quotes are synthetic.
- **Cut** (roadmap only): the nightly audit drill (P8), the lender report (WP-13), the GPU injection-classifier run and the market backtest (WP-17), the Vercel mirror (WP-18).

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4. Security design (summary)

The full self-assessment, with the generated tier table, STRIDE threat model, OWASP Top 10 for LLM Applications mapping and the known-risk list, is `docs/submission/security-self-assessment.md`. This section summarises the design. All of it is in `backend/app/security/` and is Built.

Control	Design	Code
Permission tiers	One table decides (a) the tier each action needs and (b) how far each role may take it. EXECUTE = an owner approved this exact card with a fresh TOTP code, under 24 hours ago, and the policy still allows it at execution. A card's own tier can raise the requirement, never lower it, so model-written cards cannot talk their way down.	`policy.py`
DENY list	`change_bank_details`, `add_payee`, `disable_audit`, `export_raw_pii`: refused for every role, including the owner; no executor can be registered for them.	`policy.py`, `approvals.py`
Approvals	Idempotency keys on request and execution; the approved card's SHA-256 is stored and re-checked; policy re-checked at execution. No executor is registered in this snapshot, so an executed card is recorded in the audit chain and nothing is sent and no money moves.	`approvals.py`
Roles	owner, accountant, staff, lender_viewer, compliance. Lender access exists only through the owner's recorded consent: named lender, purpose, scopes, at most 30 days, revocable, created with step-up.	`roles.py`, `team.py`
Authentication	argon2id passwords; RFC 6238 TOTP, single-use per step; server-side sessions in HttpOnly SameSite=Strict cookies, 30-minute idle and 12-hour absolute limits; 5 failures in 15 minutes lock for 15 minutes. Demo accounts only in demo mode; production refuses demo mode.	`passwords.py`, `totp.py`, `sessions.py`, `lockout.py`
Audit chain	Per-tenant SHA-256 hash chain; ORM and database triggers refuse UPDATE and DELETE; verification finds the first bad link; daily heads anchored (§3.6). Details hold ids, counts and digests only.	`audit.py`, `jobs.py`

Control	Design	Code
Injection guard	Rules for English, Bangla, Banglish and Chinese on normalised text (NFKC, case folding, zero-width and bidi controls removed, look-alike letters folded). Hits are quarantined server-side and only their SHA-256 is audited. A classifier slot exists; no classifier is installed (§7).	`guard.py`, `guard_rules.py`
Web hardening	CSP, HSTS, nosniff, frame denial, referrer and permissions policies, no-store on API data; per-route token-bucket rate limits; Origin check on state-changing requests; explicit CORS origins (wildcard refused when deployed).	`web.py`
Agent rails	Agent tools are READ to DRAFT only; anything that moves money or commits the business becomes an action card that needs a human approval. A run has a budget of tool calls, model calls and tokens. Model output never supplies a number (the checker recomputes every claim) or a tier.	`agents/tools.py`, `agents/budget.py`, `agents/checker.py` (WP-10)
Tenant isolation	Every query is scoped to the principal's tenant; other tenants' records answer 404; a `tenant_id` in a request must match the principal.	`deps.py`, routers
Secrets	Keys only in server env files and CI secrets; `.env` is git-ignored; gitleaks scans the tested commit's full history in CI.	`.github/workflows/ci.yml`

5. Data

5.1 Synthetic SME universe (labelled synthetic)

`data/synthetic/generate.py` (seed 20261020, deterministic: same seed, same bytes; file hashes in `data/synthetic/universe/manifest.json`) generates five Dhaka SMEs with 18 months of daily data, 2025-04-01 to 2026-09-30 (as-of date 2026-09-30):

SME id	Display name (fictional)	Sector
`sme_grocery`	Mirpur Mudi Store	grocery
`sme_pharmacy`	Dhanmondi Care Pharmacy	pharmacy
`sme_fabric`	Islampur Fabric House	fabric wholesale
`sme_garment`	Ashulia Stitch Works	garment subcontractor
`sme_hardware`	Nawabpur Hardware Traders	hardware

Source: `data/synthetic/universe/<sme>/profile.json`. Each shop has raw files in the formats a real shop would export (bKash, Nagad, bank statement, cash book, baki register, bills register, obligations) plus `transactions_truth.csv` as typed ground truth, so the ingest parsers are tested on realistic files. The data has weekday, month-end and festival seasonality, credit sales with per-customer delay habits, recurring rent, salary and EMIs, and the three-firm debt ring of §3.5. Every row carries `data_mode = synthetic`, and the web shows a synthetic-data banner in demo mode. Real anonymised shop data (WP-20) is not in this snapshot: field visits did not produce data in time. The approved substitute is public real data for the forecast backtest (a Bangladeshi retailer's daily sales 2013-2017, Mendeley DOI 10.17632/xwmbk7n3c8.1, and UCI Online Retail II), never described as "our shops" (`docs/packages/WP-20.md`).

5.2 Bangladesh financing catalogue (public, sourced)

`data/public/financing/bd.yaml` holds 13 priced products and 5 unpriced ones (counted with `load_catalogue()` and `load_unpriced()`). Every row was checked on 2026-10-18 and carries a `source_url` and a verbatim `evidence_quote`; the loader rejects unknown fields. Each rate has a `rate_basis`: `published`, `rate_sheet_upper` (upper bound of a published range), `regulatory_cap` (for example the Microcredit Regulatory Authority's 24% cap) or `last_published`. The UI shows the basis next to every rate. Assumptions are stated in the file header (reducing balance unless stated; Bangladesh Bank's 0.5% processing-fee ceiling applied without its Tk cap; trade licence assumed where a page lists no eligibility, labelled "typical").

Products whose provider publishes no rate (for example invoice financing, factoring and two bKash-channel loans) are listed under `unpriced` and shown as "ask the lender". **They are never ranked on a guessed cost.** Regulatory facts behind the catalogue, with quotes, are in `data/public/financing/bd_regulatory.yaml`. A second catalogue, `cn.yaml`, holds 3 products modelled on Chinese digital SME credit (P14). They are illustrative, labelled `is_modelled: true` and shown as "modelled"; they show the engine is portable, not real offers.

5.3 Other data

- Festival calendar: `data/public/calendar/bd_festivals.yaml` (Eid dates are +/- 1 day).
- Injection corpus: `data/public/injection/injection_corpus.jsonl`, 192 synthetic rows (96 attack, 96 benign) in four languages, built by `data/public/injection/build.py`.

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6. Testing and CI

Counts: see `docs/submission/evidence/test-counts.md` (regenerated by `npm run evidence`).

Layer	Tool	Where
Unit and parametrised scenario tables	pytest	`backend/tests/<area>/`
Property-based invariants	Hypothesis (derandomised in CI, `HYPOTHESIS_PROFILE=ci`)	netting invariants, graph, EAR, eligibility, fit and lender score, Monte Carlo, levers, provenance, dedupe, anonymiser round-trip, vault, contracts
Contract drift	pytest + CI step	JSON Schema vs `docs/api/contracts.json`; generated `web/src/api/contracts.ts` must not change on regeneration
Web units, i18n completeness, accessibility	Vitest, axe	`web/src/**/*.test.tsx`
End-to-end	Playwright	`web/e2e/` (smoke tests; the journey suite, WP-15, about 12 flows, is not in this snapshot)

Offline rule. Tests never call real LLMs or the network: a session-wide fixture in `backend/tests/conftest.py` makes any non-loopback socket connection fail the test. The Sepolia anchorer is tested against a fake web3 object. The live OCR test runs only where Tesseract `ben+eng` is installed (the API image), and is skipped in CI.

CI (`.github/workflows/ci.yml`, on `main` and every pull request):

1. Backend: ruff, pytest, `pip-audit --strict` on the locked dependencies.
2. Web: ESLint and TypeScript, contracts in sync, Vitest, `npm audit --audit-level=high`, build.
3. Secrets: gitleaks over the full history of the commit under test (an all-refs scan is planned before the repository is made public).
4. Docker stack: build all images, boot the Compose stack, then smoke-test through Caddy: TLS, `/api/health` reporting PostgreSQL, SPA fallback, CSP, HSTS and `X-Frame-Options` headers, HTTP to HTTPS redirect.

The evaluation harness (WP-16) runs offline with `npm run eval` and writes its step logs to the evidence bundle; it is not a CI gate. Mutation testing was cut on 19 Oct.

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7. Honest limitations

1. **Reduced or cut:** receivables lost credit limits, the dispute log and the early-pay calculator; messaging has no sending channel (the owner copies the approved text). **Cut:** the nightly audit drill and the lender report. **Market (reduced):** prices are TCB's daily Dhaka retail lists as published (10 lists, 10 to 19 Oct, saved in the repo, not fetched on a schedule), so they are a day old at best and never "live"; vendor quotes are generated for the demo shops' own ledger suppliers (synthetic) and supplier reliability is read from bill payment timing, because delivery dates are not recorded; a self-service shop cannot add quotes yet.
2. **All data is synthetic.** Backtest numbers in §3.3 and the examples in §3.8 and §3.9 come from shops our own generator produced. They test the harness, not real-world accuracy. No real shop data is in the repository.
3. **The injection classifier is not deployed.** The CPU baseline (DistilBERT multilingual, 4 epochs) reaches AUC 0.6875 on unseen attack families, with precision 0.5714 at the recall 1.0 threshold (source: `ml/artifacts/injection_metrics_cpu_baseline.json`). That is too weak to use, so the guard is rules-only. The GPU run was cut.
4. **The rule guard is proven against known attacks, not unseen ones.** The rules catch all 96 attacks and flag none of the 96 benign texts in the corpus (`backend/tests/security/test_guard.py` over `data/public/injection/injection_corpus.jsonl`), but they were written with sight of that corpus. On 12 new attacks written without reading the rules they caught 3 (the copilot still acted on none, and no approval was granted). We widened the rules per family afterwards; they now catch those 12 with no new false alarms, but that is a regression result too. On a fresh held-out set they caught 15 of 24 attacks and wrongly blocked 10 of 24 innocent questions that use the same words: the guard slows an attacker down, and the approval rules (owner plus a fresh code for anything that leaves DecisionGuard) are what keep money safe.
5. **Fit is the highest daily breach probability over the forecast, an approximation.** It uses the tenant's WP-04 band (normal approximation from P10/P90). When the riskiest day falls before the first repayment, every option gets the same fit and fit does not change the order (the synthetic grocery: 0.48% for all options). With too little history the term is dropped and options carry `fit.no_forecast`.
6. **Eligibility encodes only the conditions we could source.** For example, the Bangladesh Bank startup fund is matched on business age of at most 144 months and a trade licence, so a 6-year-old synthetic shop passes. Real lenders apply more criteria. Approval likelihood and days-to-cash are heuristics, shown as such.

7. **No lender has a track record and every licence status is `unknown`:** no verified loans are stored and no licence registry is loaded (`backend/app/api/financing.py`).
8. **On-chain anchoring is code and tests, not yet evidence,** unless the evidence index lists a Sepolia transaction. In offline mode anchors are stored locally and are not independent evidence.
9. **The model layer was measured offline.** Every copilot number in this document was produced with no model: the deterministic planner planned and narrated, and no agreement was claimed. With keys, free-tier providers impose rate limits and may train on (anonymised) inputs. Questions outside the planner's intents abstain as out of scope.
10. **Deployment.** No public hosted instance is part of this submission. The stack is verified by the CI boot test and runs locally (`npm run dev`) or with Docker Compose (`docs/DEPLOY.md`). China reachability from mainland nodes is unmeasured.
11. **Platform risks** carried from the self-assessment: in-process rate limits; no phishing-resistant second factor or recovery codes; an owner can be locked out for 15 minutes by someone who knows the email. Details: `docs/submission/security-self-assessment.md` §9.

Appendix A. Reproducing the worked examples

From `backend/` with dependencies installed (`npm run setup`). Every run is offline and deterministic (seed 20261020).

Example	How
Backtest table (§3.3)	`uv run python -m app.engines.cash.backtest --sme sme_grocery` (likewise for each SME)
Cash forecast and what-if (§3.3)	`forecast_baseline(...)` on `load_universe().smes["sme_grocery"]` as of 2026-09-30 with `load_festivals()`; `draw()` then `run()` for the plain run and for `Scenario(sales_change_pct=-10, collection_delay_days=7)` on the same draws; `buffer_target()` of the plain run; `find_gaps()`, `tier_gap()`, `smoothing_levers()` with the BD catalogue and the demo ring
True cost table (§3.4)	`true_cost(product, Money.of("50000"), 180)` for each catalogue row; `early_pay_discount_cost(Decimal(2), 10)` and `(..., 30)`
Netting ring (§3.5)	`propose_rounds(build_graph(load_universe().mesh_obligations), now=..., allow_new_links=False / True)`
Analysis (§3.8)	Seed the universe (`uv run python -m app.facts.seed`), sign in as the hardware owner, then `GET /api/analysis/statements` and `/api/analysis/ratios` with `granularity=quarter&on=2026-09-30`, `POST /api/analysis/scenarios` with `{}`, and `POST /api/analysis/stability` with the returned `recommendation_id`
Copilot (§3.9)	Same seeded database, `LLM_MODE=fixtures` and no keys: `POST /api/copilot/ask` with `{"question": "Will I run out of cash in the next month?"}` and `{"question": "How big a loan can I safely take?"}`
Guard regression	`uv run pytest tests/security/test_guard.py -q`

Through the API, the same cash numbers come from `GET /api/cash/forecast` and `POST /api/cash/whatif` for a signed-in demo owner of `sme_grocery` (demo mode). The operation log (`docs/submission/evidence/operations/operation-log.md`, regenerated by `scripts/evidence_operations.py`) records 48 such API and engine steps end to end.